

Subject Wise Motor Imagery Classification from EEG Data Using Transfer Learning

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Abstract—Machine learning (ML) has no doubt virtually helped in nearly all fields of life, including medical sciences. ML models are now being trained, tested and developed with the help of information gained from Electroencephalogram (EEG) Signals. Neural Networks (NN) are being used specifically in this regard to exploit their image classification ability. A special class of NN called Transfer Learning (TL), is used to enhance the capability of NNs. In this paper, EEG signals are extracted and used to classify Left or Right Motor Images of the brain using InceptionV3 and VGG16 models. We try to enhance the accuracy of these TL Models by exploiting a different methodology as compared to other available statistical methods available in the research community. For the said purpose, a dataset from Brain-Computer Interface (BCI) Competition IV 2b was used. EEG signals are extracted and transformed into Short Time Fourier Transform (STFT) images. These STFT images are labeled with either Left or Right Motor Imagery (MI) Class. The transfer learning models are trained using these STFT images and results are also compared with a state-of-the art research, implementing Capsule Networks.

Index Terms—EEG, Transfer Learning, BCI

I. INTRODUCTION

Classifying Motor Images from the Left and Right parts of the brain is a gruesome task, even with the most sophisticated neural networks. Brain-Computer interface (BCI) is one of most elaborate technologies to have emerged. Recently, works have been done focusing on experimenting with BCI competitions, and applying ML techniques in this regard to uncover the hidden structures or information in these EEG data. Only a few selected papers can be found regarding use of transfer learning in EEG signaling. For example, [1] used transfer learning to perform multi-subject classification. Today, a lot of tasks are being implemented in deep learning. A number of new algorithms and models are trained and tested, specifically in the medical field. Classifying Motor Imagery data from EEG is a bit challenging as the STFT images obtained do not have any such distinctive features for the model to accurately train and dataset is also not large enough.

Transfer learning is a new paradigm in machine learning where weights of neurons in a Neural Network (NN) are already pre-trained. This means, we do not have to train the entire model again for our usage. We can simply import the weights and network itself, and with a few adjustments, can

re-design the NN according to our requirement. We simply have to adjust the input size of the network, according to our image. A number of additional layers can be added to increase the performance of the NN.

Convolutional Neural Networks (CNN) are also available today that give a reasonable amount of accuracy. A point worth mentioning is that STFT itself does not any distinctive feature. This makes it hard to train and test CNN. To train STFT images on CNN, a lot of layers will be added to increase the performance. The issue here is the training of these layers. Computational complexity may increase significantly. This where Transfer Learning comes into play.

In this paper, we propose a non-statistical approach to produce STFT images and used them to train and test the TL model. The proposed work is also compared with the research work carried out in [2], which implements Capsule Networks. We also avoid any augmentation technique in order to analyze the behaviour of the dataset with the original size.

Section II presents Literature Review. Section III discusses the details about the dataset. Section IV gives an insight about Transfer Learning, whereas Section V briefly discusses the Implemented Methodology along with the results. The experimental setup and results are discussed in Section VI. Conclusion of the proposed methodology is presented in Section VII.

II. LITERATURE REVIEW

BCI has really provided a breakthrough in terms of what we can achieve as humans. Although recording of EEG signals has been a huge milestone, we pushed the limits with the help of machine learning. With ML, we can actually take these EEG signals and perform various classification procedures. Neural networks are now being applied in this field to help classify images from various medical fields. Because of the accuracy a neural network can provide, usually a huge number of images is required for testing and training. In this regard, achieving accuracy greater than 90 % is not very much achievable among the research community. For instance, authors of [3] have used EEGnet with average accuracy of 66.36% whereas research in [4] has also used transfer learning with not much

of improvement as compared to not using transfer modeling, given the nature of the problem.

EEG signals have been statistically explored among academics and researchers. A common transformation technique, called Common Spatial Patterns (CSP) algorithm is mostly used with variations. The problem with CSP is that the frequency band of the EEG signals have to be preselected, which can create extra load and possibly decrease the efficiency of the overall model.

A number of works are present which make use of the CSP algorithm. For example, [5] has used CSP for feature extraction and LDA for classification analysis. They achieved an accuracy of about 70%. Authors of [3] have used another technique called augmentation. They have used a smaller number of nodes in CNN, leading to less weights to be trained. They augmented the EEG data and improved its accuracy. Similar, a most recent work has been done in [6] where they enhanced algorithms from classical machine learning to multi-layer perceptrons. An overall review in EEG classification is given in [7]. It can be observed that most of EEG classification work has been done using Deep Learning. There is need to explore more insight of using transfer learning in EEG classifications. Other methods include Fourier Transform, Wavelet Transform and Hilbert Transform. A detailed inspection of use of EEG in Neural Networks is carried out in [8].

Authors of [9] proposed an improved algorithm, combining an energy-based feature (of the brain) and SVM, and analyzing the final results using Wilcoxon signed-rank test. They apply Filter-bank CSP (FBCSP) to preprocess the BCI IV-2a dataset. They have been able to classify Left and Right MI with the optimized methodology. Similarly, using the BCI IV-2b dataset, [10] obtained the frequency representations of EEG signals using the Continuous Wavelet Transform (CWT) and used it to train a CNN model. They extracted 2 seconds of each EEG segment and used two different mother wavelets, Morlet and Bump.

A similar, yet effective methodology is proposed in [11]. Authors used 12 electrodes from the available 22 electrodes in the original dataset. They discard EOG signals and use only EEG signals to train a CNN and a LSTM model and able to classify Left and Right MIs. They argue that reducing the number of electrodes may or may not reduce the overall accuracy. Depending upon the extraction methodology, one can achieve higher or even near-competing accuracy to reduce the overall complexity of the model.

Researchers in [12] have used AlexNet, ResNet50 and InceptionV3 CNN models to classify Left and Right MI EEG Signals. Out of the three, InceptionV3 gives the highest accuracy. They conclude InceptionV3 can be used in the EEG domain. During the whole process of EEG processing, features are usually hand-crafted, which makes the entire process computationally expensive. But on the bright side, a well good range of accuracy can be achieved. Authors of [13] have constructed a Long-Short Term Memory (LSTM) without the use of hand-crafted features. The model also learns to filter the input automatically without user intervention, which makes

the whole process simple.

In [14], authors transformed the EEG signals to images based on the location of scalp electrodes. They tested CNN using 2D and 3D kernels on the EEG data collected on campus. As compared to the traditional FBCSP method implemented in [9], authors have proved that their implemented Neural network model performs better. They concluded that their CNN model classified Left and Right MIs with an accuracy of $68.38 \pm 7.29\%$ and $65.94 \pm 8.52\%$ with 2D and 3D kernels respectively.

In all of the above mentioned works, they accurately classify the Left and Right MI of the brain. Above research works focused on some sort of statistical method to enhance the EEG at first. These somewhat filtered EEG signals are then fed to a NN. Depending upon the extent of research, statistical methods on EEG can somewhat be simple to complex problems. Complex statistical methods such as Hilbert Transform can be computationally expensive. In our research, we specifically focus only on EEG signals without any statistical method, thus reducing additional complexity. EEG signals are transformed into STFT images, which are fed to the transfer learning model.

III. DATASET

For this experiment, the dataset from BCI Competition IV 2b was used. 9 subjects were recruited for the process. The experimentation was divided into 2 parts. The first, providing a cue without any feedback. The second, providing the cue and recording the feedback. EEG signals for both parts were recorded by sensors and provided in the dataset. Detailed explanation of the dataset along with the entire experiment is provided by the researchers in [15].

A total of 6 signals were recorded, 3 EOG and 3 EEG. For our purpose, we will ignore EOG signals and simply keep the focus on EEG signals. The sampling frequency of the recorded signals is 250 Hz. Extracting 2 sec of each event i.e. 769 & 770, gives us 500 samples/events. This information is used later on to calculate STFT of images in the subsequent sections. There were a total of 9 subjects who participated in the experiment. 5 files were constructed against each of the 9 subjects. Out of those 5 files, only 3 files were used in our proposed model. The remaining 2 were discarded i.e. files ending with E due to difficulty in extracting the required events.

IV. TRANSFER LEARNING

By intuition, Transfer Learning means to use the knowledge gained in one domain, in another related domain. With respect to the research carried out in this paper, Transfer Learning is the use of a pre-trained Neural Network and implementing it in another problem where relevance can be achieved to some extent. Today, Neural Networks are mostly used in Digital Imaging where classification of images is done to achieve a variety of tasks. For the said task, more the layers of a NN, more accurately it can classify images.

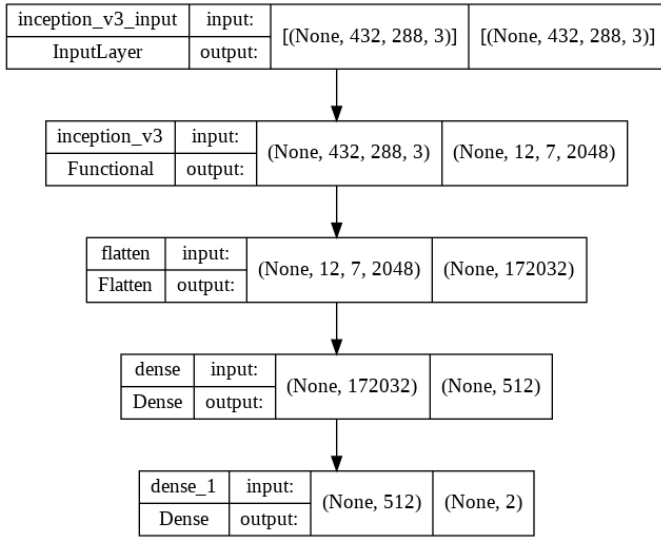


Figure 1 Proposed InceptionV3 Model

Authors in [16] conducted a research to use a 5-layer CNN to classify Left and Right MIs. Unlike the methodology proposed in this research, they have used augmentation techniques to enhance the size of the dataset. Exponential Linear Function (ELU) is used as the activation function. As Independent Component Analysis (ICA) is unable to fully remove EOG artifacts from EEG, they use regression based correction method. It is a linear algebra based method to filter out EOG from EEG signals. Although, the authors carried out good enough research, there is potential to enhance the methodology to reduce the mathematical processes and achieve near-about results.

Although some thinking needs to be devoted to choose the appropriate NN algorithm to classify Left and Right MIs, choosing the right algorithm for the right purpose can still be hectic. For this purpose, research carried out in [17] provides a base model for choosing an algorithm for a specific purpose. They studied various Riemannian transfer learning algorithms keeping in view a few considerations. Both supervised and unsupervised algorithms can be used for MI classification. Alongside transfer learning, tensor based classifiers are also being used for EEG MIs. For an extensive survey on the use of supervised, unsupervised, transfer and deep learning algorithms in EEG signals, specifically MI classification, an excellent survey is presented by authors of [18]. They also reviewed feature selection and extraction techniques with specific regard to EEG signalling. They surveyed research articles from 2007 to 2017, and concluded their work with future gaps. It is clear that CNN, when combined with various statistical techniques, can produce accurate results [18].

Generating a complex NN does solve the problem of classifying images but creates a new problem as well. A complex NN will take up a lot of computational power; resulting in more processing power and time consumption. For this purpose, today's complex NN are mostly trained and used on

Graphical Processing Units (GPUs). Training a complex model can take days if not weeks to train. To solve this problem, researchers from around the world working on complex NNs create and pre-train the weights and make it available as an open-source model for the academic community. A number of models are available today in Python's Keras library. VGG16, AlexNet, InceptionV3, InceptionV6, ResNet50 are among the mostly used transfer models because of their performance and adaptability to most problems regarding classification of images.

A. VGG16

VGG16 has 16 layers having trainable parameters. Overall, it has 13 convolutional layers, 5 Max-pooling layers and 3 Fully-connected layers. By adding all trainable parameters, VGG16 has over 138 million parameters. Training this amount of parameters is a matter of weeks on traditional processing processors. This is where the concept of Transfer Learning comes into play. All hidden layers are activated by Linear Rectifier (ReLU) function. We simply import the Neural Network with pre-trained weights, adjust the input by adding an additional layer, and execute the model. Overall information of parameters of VGG16 implemented in this paper is as follows:

- Trainable Parameters: 30,672,386
- Non-trainable Parameters: 14,714,688
- Total Parameters: 45,387,074

B. InceptionV3

InceptionV3 along with VGG16 is used for image classification. It has a total of 42 layers. It is computationally less expensive than its predecessors i.e. InceptionV1 and InceptionV2 and also has higher efficiency. InceptionV3 is a combination of convolutional, pooling layers and modules from its older V1 and V2 models. InceptionV3 model summary carried out in this research is shown in Figure 1 whereas number of parameters is given as follows:

- Trainable Parameters: 88,081,922
- Non-trainable Parameters: 21,802,784
- Total Parameters: 109,884,706

V. PROPOSED METHODOLOGY

The step-by-step procedure for implementing the proposed algorithm is given below. Each of the steps is briefly explained as well. The process is also depicted in Figure 2.

- 1) Extraction of raw data from .gdf files.
- 2) Data processing to form a dataset consisting of ONLY Class1 (Label 769) and Class2 (label 770).
- 3) Performing STFT on each data sample of the extracted events.
- 4) Manually associating labels with each of the STFT images
- 5) Importing and training the model using above STFT images
- 6) Evaluating the model using testing files using `model.evaluate()`

A. Extraction of Features

- 1) Reading the raw data from .gdf files along with specifically mentioning the EOG channels to exclude them later on.
- 2) Preprocessing the extracted data and applying ICA filter in order to remove any data correlations (if any).
- 3) Extracting the events i.e., 769 & 770 from the RAW data
- 4) Finally, calculating Epochs for the events.

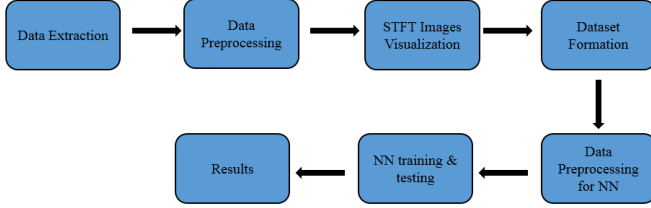


Figure 2 Methodology

B. Preprocessing

Originally, each EEG contains 13 individual events which are annotated within the signal. For motor imagery classification, we require only 2 classes, Left and Right Motor cortex's. These two classes are labeled 769 and 770 respectively by the providers of the dataset. Also, 3 EOG channels are also ignored for this purpose to carry out training purely on EEG images. After dropping the EOG channels and picking only events belonging from class 769 and 770, the data is now ready to be analyzed in them respective STFT images. Each one of the EEG channels further has 3 electrodes, from where data is being obtained. The electrodes are named C3, Cz and C4.

C. Applying STFT

Once data has been cleaned out, it is now fed to produce their STFT images. Python's `signal.stft()` command is used to calculate the image. Sampling frequency of 250 Hz is provided to accurately calculate the STFT of the processed signal. As mentioned in Section III, Spectrogram is calculated for 2 seconds. Depending on the number of channels taken, the number of STFT images can vary. For instance, 3 files against 1 subject was taken. Each spectrogram reflects each channel of the each of subject. For instance, spectrogram obtained from channel 1 (out of three channels) of subject 1's 1st file (out of 3 files) is depicted in Figure 3.

D. Label Association

STFT images do not have labels (i.e. class 769 and 770) assigned to them, so these labels have to be assigned manually. From an implementation view, images of STFT are kept in separate folders and assigned labels according to all images in a single folder i.e. images in folder class 769 are assigned (for instance) label 0 and class 770 are assigned label 1. Step 5 and 6 i.e. training of neural network along with model evaluation is discussed in the subsequent section.

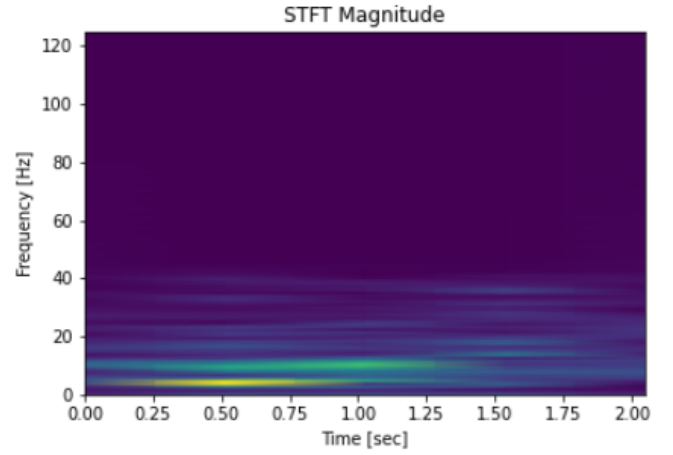
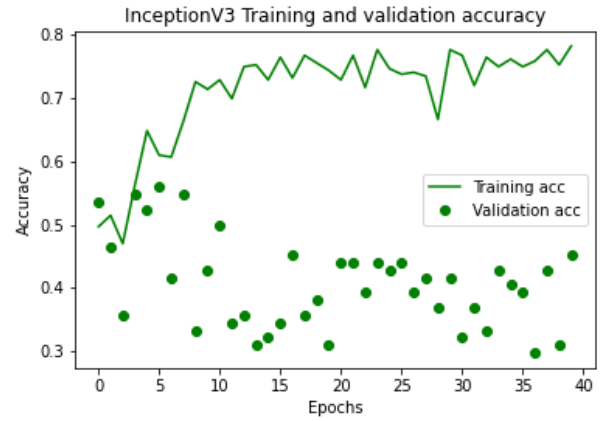
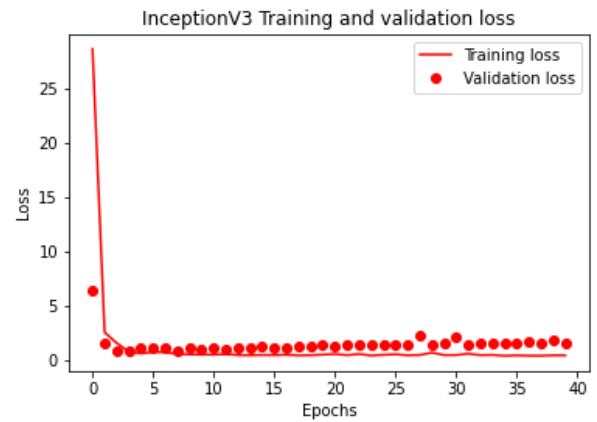


Figure 3 Spectrogram obtained from STFT



(a) Training & Validation accuracy



(b) Training & Validation loss

Figure 4 Training & Validation Accuracy & Loss for Subject 6

VI. EXPERIMENTAL SETUP AND RESULTS

For experimentation, Google COLAB was used to carry out the NN testing. Instead of training on the network on a traditional CPU, which would take a significant amount

```
raw=mne.io.read_raw_gdf(path,eog=['EOG:ch01','EOG:ch02','EOG:ch03'], preload = True)
ica=mne.preprocessing.ICA()
ica.fit(raw.filter(4,38))
event,eventid=mne.events_from_annotations(raw)
event_ids={'769':eventid['769'],'770':eventid['770']}
epochs=mne.Epochs(raw,event,event_id=event_ids,tmin=-.5, tmax=1.5,preload=True)
```

Figure 5 Event Extraction from EEG data

of time, GPU setting was enabled on COLAB to speed up the training and testing. A total of 30 epochs were run, which took about 7~8sec/epoch. The steps given in Section V are implemented step by step. Brief explanations of the implemented steps is discussed below.

Step 1 The process of extracting events from gdf files is shown in Figure 5.

Step 2 Preprocessing

After STFT images are obtained, their labels are assigned and combined in a single dataset to be fed to the NN. The data has to be preprocessed according to the transfer learning models. Each Transfer Learning model has its own command to be used. In case of InceptionV3, preprocessing of the dataset X is achieved by using the builtin keras command depicted in Figure 6. Whereas, the prediction feature i.e (Left or Right MI EEG) has to be converted to *categorical* values from *numerical* values (in Step 1), so they can be used in the algorithm.

```
X_p = K.applications.inception_v3.preprocess_input(X)
Y_p = K.utils.to_categorical(Y)
```

Figure 6 InceptionV3 Preprocessing

Step 3 Importing Transfer Learning model

Subject wise training and testing is performed on the NN. Each model is trained from (say) Subject 1 and then tested on the same dataset. Before feeding the STFT images to the neural network, respective labels have to be assigned to them. A sample STFT spectrogram is shown in Figure 3. After assignment of labels, data has to be preprocessed, specifically for InceptionV3 model. The InceptionV3 model is imported and the weights are made non-trainable. Input is adjusted to 432x288 which is precisely the size of the STFT images. Each individual model was run for 30 epochs.

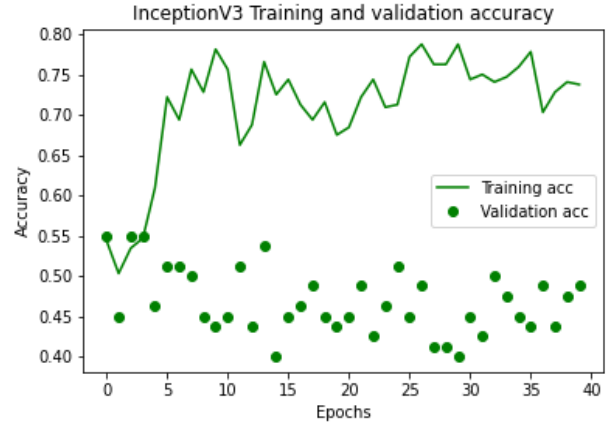
Table I COMPARISON OF ACCURACIES OF [2] & PROPOSED ARCHITECTURE

Subject	Accuracies of [2] (%)	InceptionV3	VGG19
1	78.75	70.25	72.50
2	55.71	58.50	50.13
3	55.00	72.50	74.50
4	95.93	55.95	74.04
5	83.12	71.00	71.98
6	83.43	72.25	73.37
7	75.62	73.00	71.88
8	91.25	66.82	69.55
9	87.18	75.00	75.00

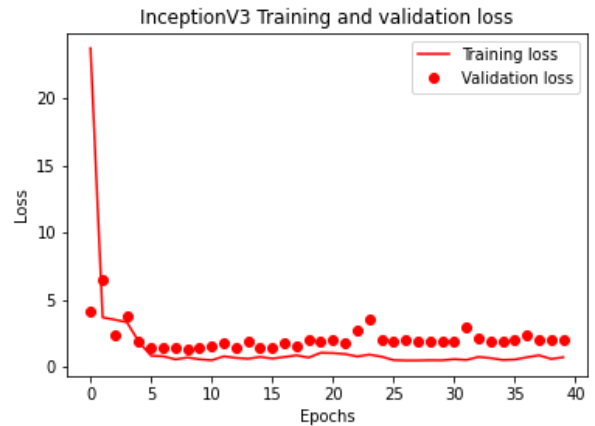
Step 4 NN Training

Table II AVERAGE ACCURACIES OF PROPOSED TRANSFER LEARNING MODELS

Model	Accuracy (%)
VGG16	68.36
InceptionV3	70.32



(a) Training & Validation accuracy



(b) Training & Validation loss

Figure 7 Training & Validation Accuracy & Loss for Subject 7

Once the initial model is loaded and ready to use, additional layers will be added to support our model. Only these layers will be trained and weights will be optimized accordingly. As shown in the Figure 1, Dense layer of 512 nodes is added to actually compress out previous Flatten layer. At the end, a dense layer is added to compress the images into 2 nodes. Each node belonging to each output class label (i.e. class 769 or class 770). After Applying CNN-based transfer learning, results clearly show that the accuracy of the InceptionV3 produces better results than VGG16 and Inception V3. Every time the model was trained from scratch for each subject. Training & Validation Accuracy and Loss were recorded for each particular subject. Subject 6 and 7 are shown in Figure 4 & 7 respectively. Whereas the accuracies recorded for each individual subject are populated in Table I.

We can observe a lot of variation in the training and validation sets. This is normally expected in EEG classifications. As we do not have solid images for the model to train on, the neural network tries hard to classify the image based on the best features it can detect. The average accuracies of InceptionV3 and VGG16 models is given in Table II.

VII. CONCLUSION & FUTURE WORK

In this paper, we made use of EEG signals by a non-statistical approach. Concerned events i.e., Left and Right Motor Cortex Images are extracted to form a new dataset along with their labels. These events are then used to calculate their respective STFT images which are then fed to the NN. The NN was trained and tested using these STFT images upon which results are provided. As obvious, given the STFT images, there are not any distinguishing features that the NN can train upon. Due to this, the accuracy could not exceed 80% in any individual case. For future cases, we will be focusing on enhancing the dataset by applying any augmentation technique to increase the size of the dataset without disturbing the original content of the data. Along with augmentation, any minimal inclusion of statistical method will also be focused.

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